

# Lab Report: SAR ADC Components

Lab Report

February 10, 2026

## SAR Logic Design and Verification

### 1 Transient Simulation Results (SAR Logic)

This section reports the transient simulation results for the SAR logic control block. The waveforms for the Ring Counter output, the Code Register output (digital word), and the End of Conversion (EOC) signal are presented for three specific comparator conditions.

#### 1.1 Case A: CMP is All Zeros

In this test case, the comparator output is held at logic '0'. This simulates a condition where the analog input is consistently lower than the DAC trial voltage. The SAR logic correctly discards every trial bit, resulting in a final digital code of 00000000.

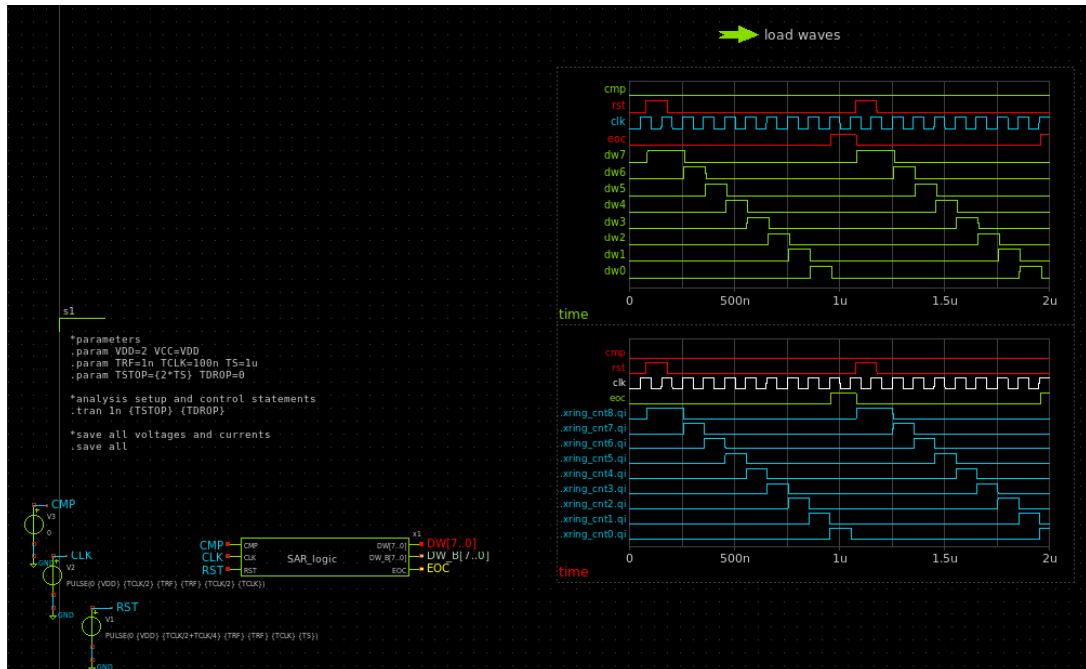


Figure 1: Simulation Results: CMP = 0. The digital word (DW) remains 0 throughout the conversion. EOC goes high at the end.

## 1.2 Case B: CMP is All Ones

In this test case, the comparator output is held at logic '1'. The SAR logic retains every trial bit, resulting in a final digital code of 11111111.

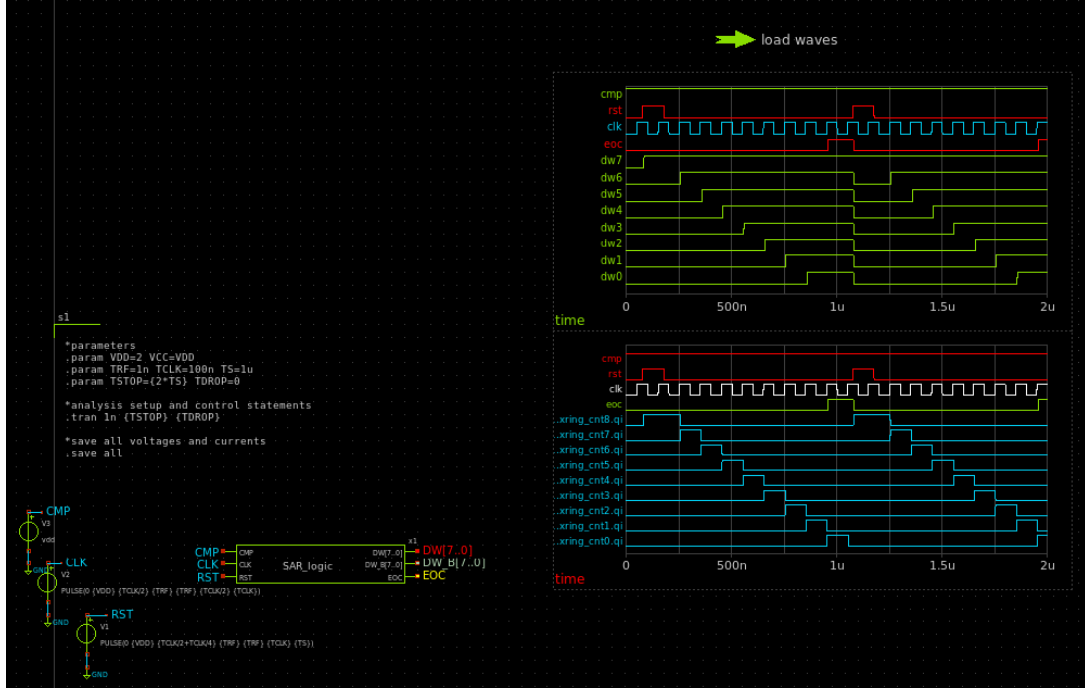


Figure 2: Simulation Results:  $CMP = 1$ . The digital word (DW) accumulates 1s sequentially from MSB to LSB.

## 1.3 Case C: CMP is Alternating

In this test case, the comparator output toggles between '1' and '0'. This verifies the dynamic latching capability of the circuit.

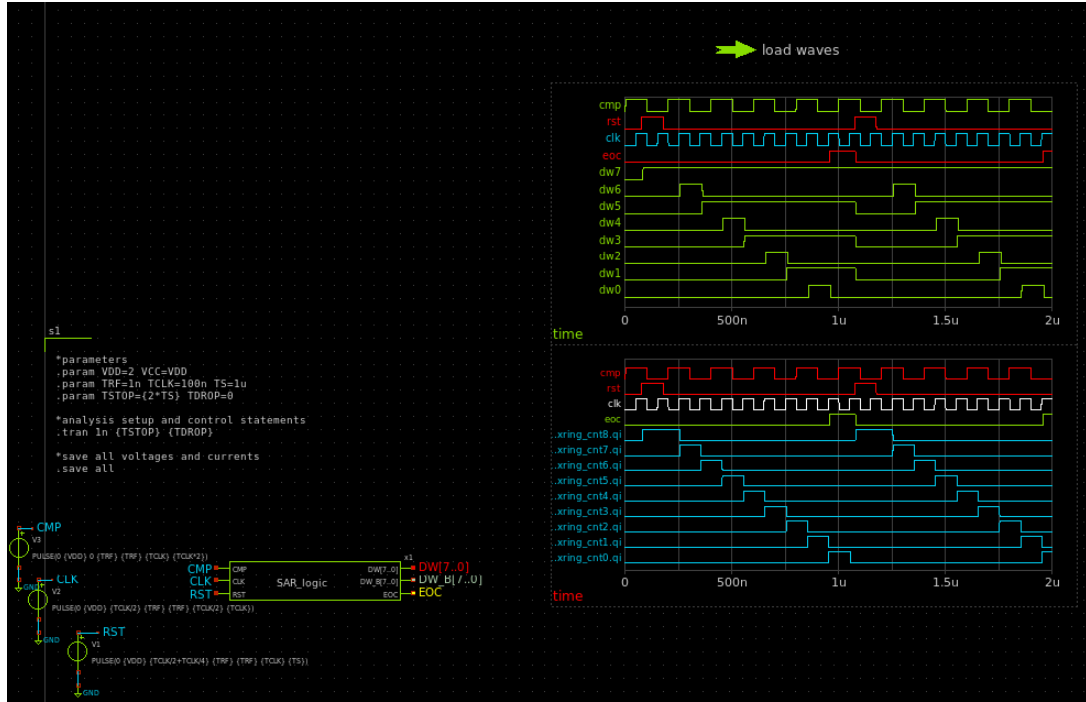


Figure 3: Simulation Results: Alternating CMP. The digital word resolves to an alternating pattern (10101010).

## 2 Design Explanation (SAR Logic)

### 2.1 Successive Approximation Algorithm Implementation

The designed circuit translates the binary search algorithm employed in SAR ADCs. The logic controller is divided into two main parts:

1. **Ring Counter (XRING\_CNT):** This is a sequencer. It starts with the MSB set and moves the "active token" one position to the right with each clock cycle. This shows which bit is in the "Trial Phase" at any given time.
2. **Code Register (XCODE\_REG):** This holds the approximation of the digital code at any time.
  - **Set/Trial:** If the Ring Counter is indicating a bit, the Code Register forces that bit High.
  - **Latch/Resolve:** On the next clock cycle, if CM P = 1, the bit stays High. If CM P = 0, the bit goes Low.

### 2.2 State Table

The following table details the operation over 10 clock cycles.  $B_x$  indicates the resolved value determined by the comparator.

Table 1: SAR Logic State Table

| Clock     | DW<7> | DW<6> | DW<5> | DW<4> | DW<3> | DW<2> | DW<1> | DW<0> | CMP   |
|-----------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1 (Reset) | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | -     |
| 2         | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | $B_7$ |
| 3         | $B_7$ | 1     | 0     | 0     | 0     | 0     | 0     | 0     | $B_6$ |
| 4         | $B_7$ | $B_6$ | 1     | 0     | 0     | 0     | 0     | 0     | $B_5$ |
| 5         | $B_7$ | $B_6$ | $B_5$ | 1     | 0     | 0     | 0     | 0     | $B_4$ |
| 6         | $B_7$ | $B_6$ | $B_5$ | $B_4$ | 1     | 0     | 0     | 0     | $B_3$ |
| 7         | $B_7$ | $B_6$ | $B_5$ | $B_4$ | $B_3$ | 1     | 0     | 0     | $B_2$ |
| 8         | $B_7$ | $B_6$ | $B_5$ | $B_4$ | $B_3$ | $B_2$ | 1     | 0     | $B_1$ |
| 9         | $B_7$ | $B_6$ | $B_5$ | $B_4$ | $B_3$ | $B_2$ | $B_1$ | 1     | $B_0$ |
| 10 (EOC)  | $B_7$ | $B_6$ | $B_5$ | $B_4$ | $B_3$ | $B_2$ | $B_1$ | $B_0$ | Done  |

# Transmission Gate

## 3 Transmission Gate Design

### 3.1 Schematic Design

A CMOS Transmission Gate (TG) was designed using parallel NMOS ( $W = 5\mu$ ) and PMOS ( $W = 10\mu$ ) transistors. This allows the gate to pass full logic levels ( $0V$  to  $VDD$ ) with minimal resistance. An ideal inverter generates the complementary enable signal for the PMOS device.

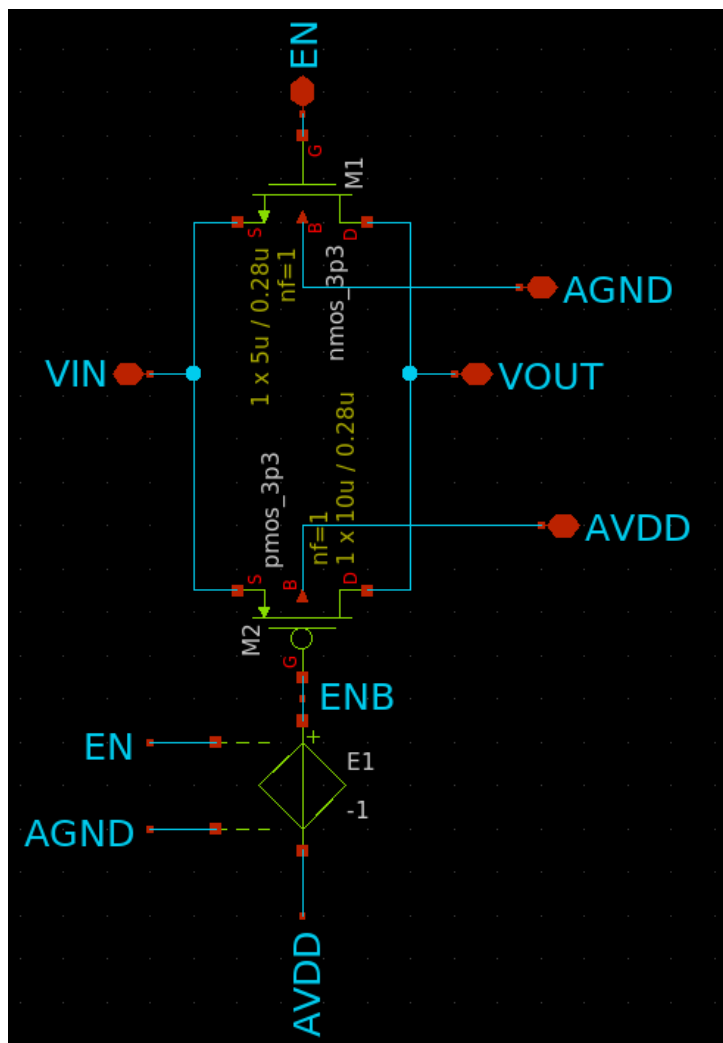


Figure 4: Schematic of the CMOS Transmission Gate.

### 3.2 Transient Simulation

The simulation verifies that the TG passes the input signal ( $V_{IN}$ ) to the output ( $V_{OUT}$ ) when enabled ( $EN = 1$ ) and blocks it when disabled ( $EN = 0$ ).

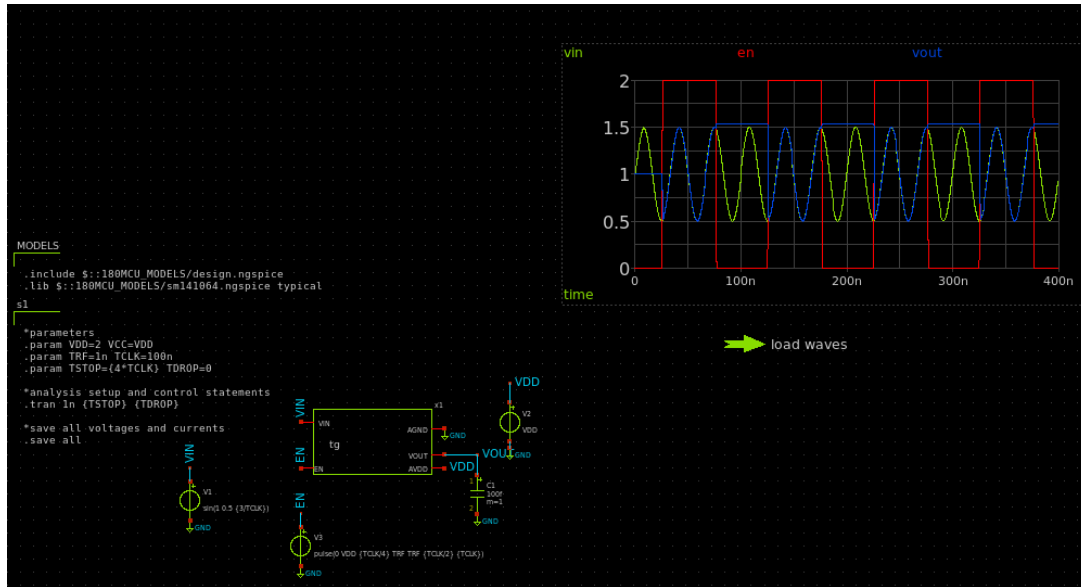


Figure 5: Transient response: Output (Blue) follows Input (Green) when Enable (Red) is high.

# Bottom-Plate Switch

## 4 Bottom-Plate Switch Design

### 4.1 Schematic Design

The Bottom-Plate Switch is a critical component in the SAR ADC capacitive DAC. It has two main modes of operation controlled by the sampling signal ( $SMPL$ ):

1. **Sampling Phase** ( $SMPL = 1$ ): The Transmission Gate ('tg') is enabled, connecting the analog input ( $V_{IN}$ ) to the bottom plate of the capacitor ( $V_{BOT}$ ). During this phase, the switches to  $V_{REFP}$  and  $V_{REFN}$  are forced OFF.
2. **Conversion/Redistribution Phase** ( $SMPL = 0$ ): The Transmission Gate turns OFF, disconnecting the input. The node  $V_{BOT}$  is then connected to either the positive reference ( $V_{REFP}$ ) or the negative reference ( $V_{REFN}$ ) based on the data bit ( $D$ ).
  - If  $D = 1$ : The PMOS switch (M1) is turned ON via the NAND gate logic, pulling  $V_{BOT}$  to  $V_{REFP}$  (1.5V).
  - If  $D = 0$ : The NMOS switch (M2) is turned ON via the NOR gate logic, pulling  $V_{BOT}$  to  $V_{REFN}$  (0.5V).

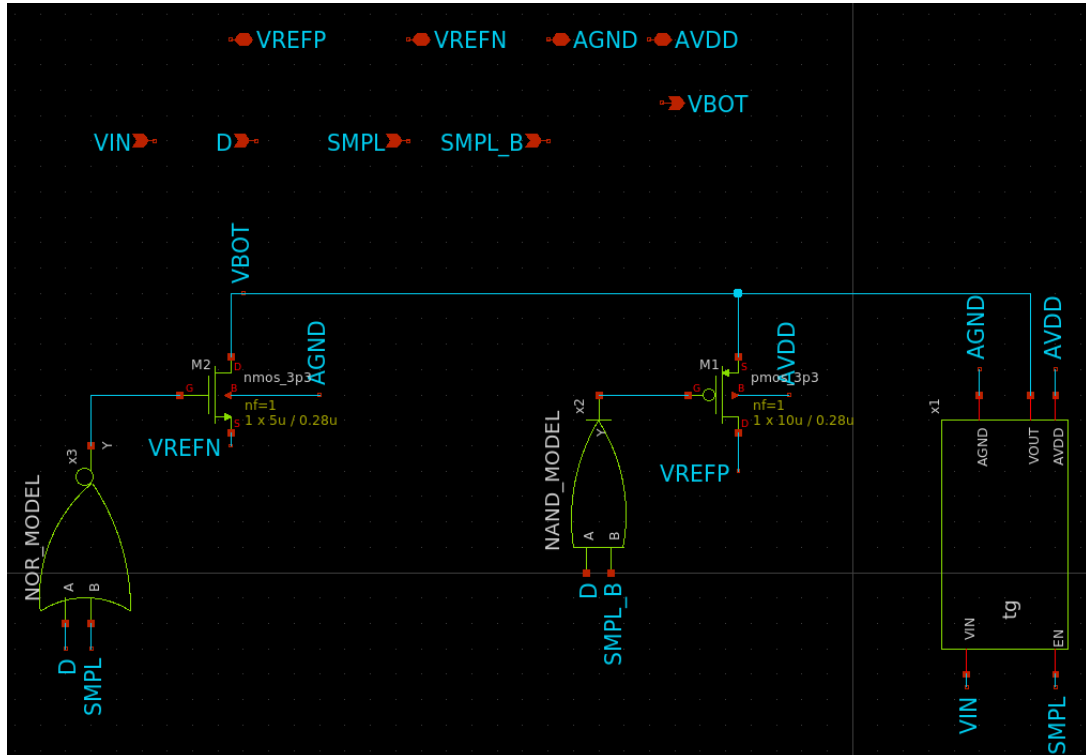


Figure 6: Schematic of the Bottom-Plate Switch. It integrates the Transmission Gate from Part 3 with logic-controlled reference switches.

## 4.2 Transient Simulation Results

The simulation below verifies both operating modes.  $V_{REFP}$  is set to  $1.5V$  and  $V_{REFN}$  to  $0.5V$ .

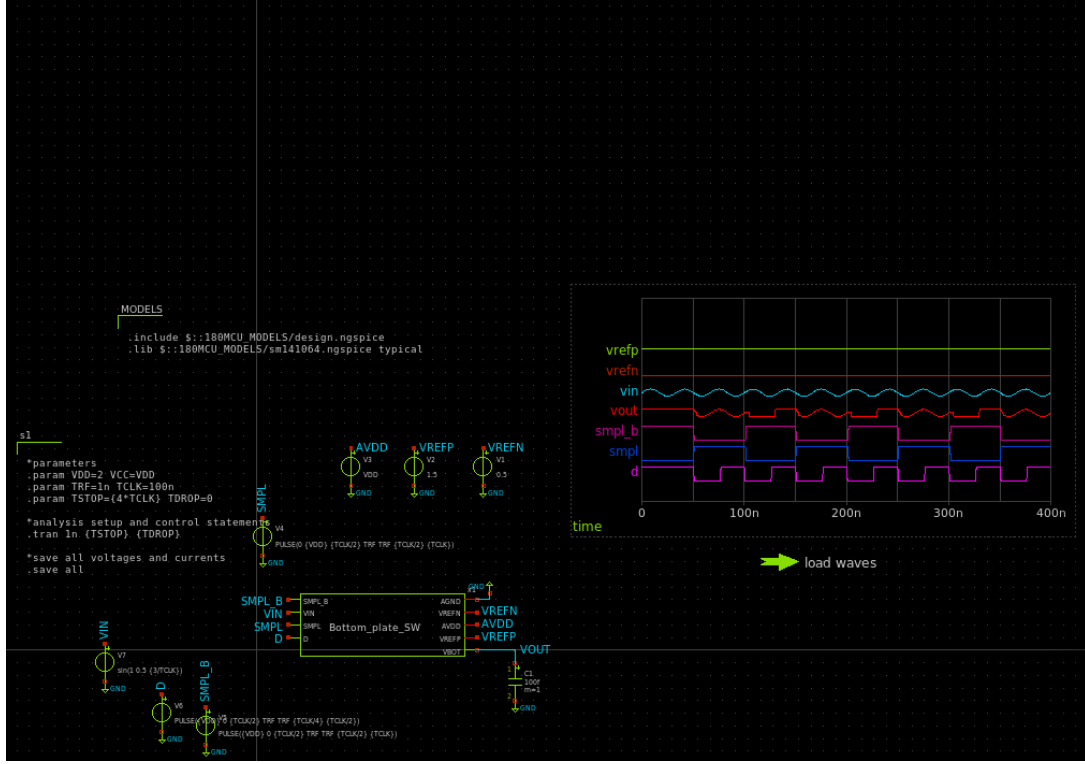


Figure 7: Transient response of the Bottom-Plate Switch.

## 4.3 Analysis of Results

The simulation waveforms in Figure 7 demonstrate correct operation:

- **Sampling (Blue Trace  $SMPL = 2V$ ):** When  $SMPL$  is high, the output voltage ( $V_{OUT}$ , red trace) perfectly tracks the sine wave input ( $V_{IN}$ , cyan trace). This confirms the Transmission Gate is conducting.
- **Hold/Convert (Blue Trace  $SMPL = 0V$ ):** When  $SMPL$  goes low, the output stops tracking the input and switches to a reference level determined by the data bit  $D$  (magenta trace).
  - When  $D = 0$  (Low),  $V_{OUT}$  settles to  $V_{REFN}$  ( $0.5V$ ).
  - When  $D = 1$  (High),  $V_{OUT}$  settles to  $V_{REFP}$  ( $1.5V$ ).

This confirms that the switch successfully toggles between input sampling and reference voltage selection, which is the required behavior for a SAR ADC capacitor array driver.



## 5 SAR ADC System Integration

## 5.1 Full System Verification

The complete SAR ADC system was integrated, connecting the Capacitive DAC ('Cap\_DAC'), the SAR Logic ('ADC\_10'), and the Comparator. The system was verified using transient simulation and FFT analysis.

## 5.2 Transient Simulation

Figure 8 shows the transient response. The waveforms illustrate the operation of the Capacitive DAC ('vdac', green trace) as it tracks the input signal ('vin', red trace) during the sampling phase and performs bit trials during the conversion phase.

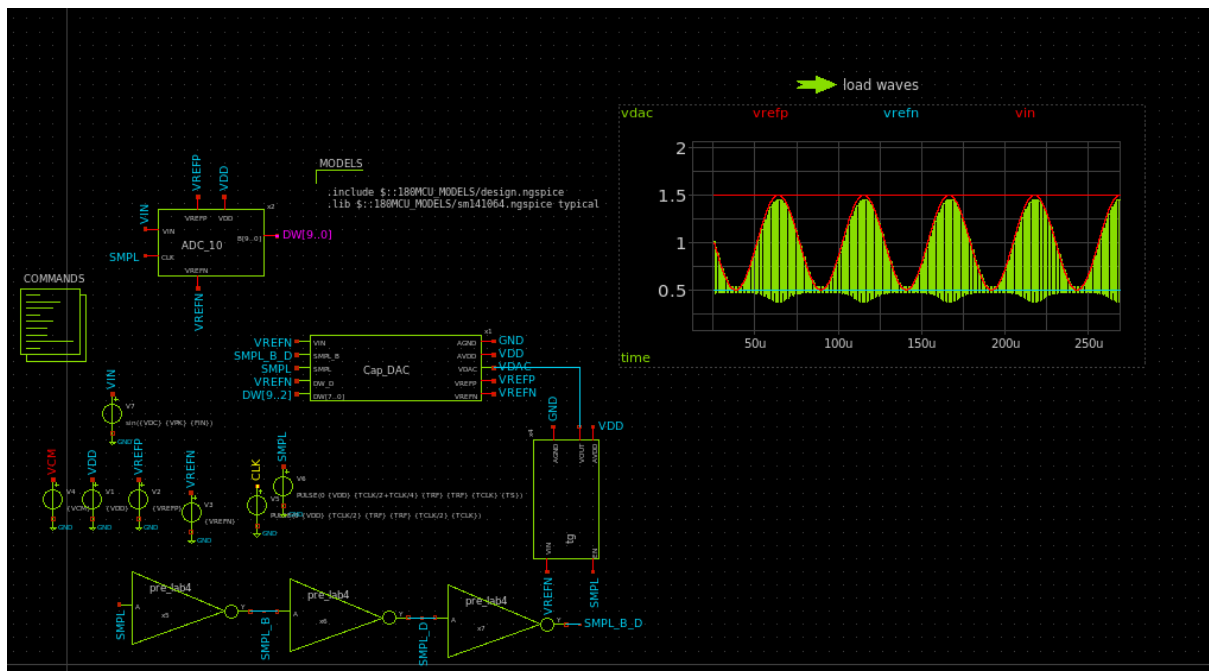


Figure 8: Integrated ADC Transient Simulation. Left: Testbench schematic. Right: Waveforms showing the DAC output (green) tracking the input (red/cyan) and then redistributing charge during conversion.

### 5.3 Signal Attenuation Analysis

**Observation:** The amplitude of the reconstructed sine wave and the DAC output is slightly lower than the input sine wave amplitude ( $A \approx -6.2$  dBFS).

**Explanation:** This attenuation is caused by **\*\*Parasitic Capacitance\*\*** at the top plate of the capacitor array (the comparator input node). Ideally, charge conservation ensures the voltage remains constant. However, parasitic capacitance ( $C_{par}$ ) from the comparator input transistors and interconnects creates a capacitive voltage divider with the array capacitance ( $C_{total}$ ).

$$V_{top} = V_{in} \cdot \frac{C_{total}}{C_{total} + C_{par}}$$

Since  $C_{par}$  is non-zero, the ratio is less than 1, resulting in a **\*\*gain error\*\*** where the signal amplitude is attenuated.

## 5.4 Dynamic Performance (FFT Analysis)

Figure 9 presents the FFT spectrum of the ADC output. This analysis quantifies the dynamic performance metrics of the design.

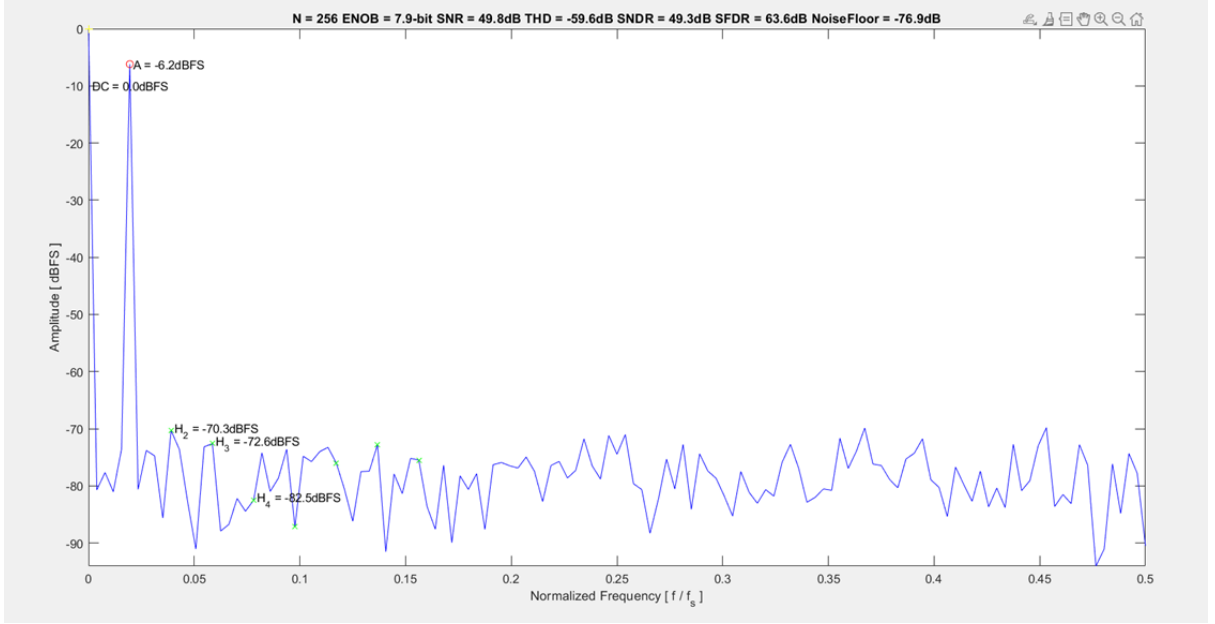


Figure 9: FFT Spectrum of the ADC output showing key performance metrics.

Table 2: Extracted Dynamic Performance Metrics

| Metric                                | Value    |
|---------------------------------------|----------|
| Sampling Points (N)                   | 256      |
| Signal-to-Noise Ratio (SNR)           | 49.8 dB  |
| Signal-to-Noise-and-Distortion (SNDR) | 49.3 dB  |
| Spurious-Free Dynamic Range (SFDR)    | 63.6 dB  |
| Total Harmonic Distortion (THD)       | -59.6 dB |
| Effective Number of Bits (ENOB)       | 7.9 bits |

## Charge Redistribution Verification

## 6 System Verification

In this part, the SAR Logic was connected to the Capacitive DAC and Comparator to verify the Charge Redistribution and Binary Search algorithm. Three distinct test cases were simulated to observe the convergence of the DAC voltage ( $V_{SAR}$ ) towards the sampled input voltage ( $V_{SAH}$ ).

## 6.1 Case 1: Low Input Voltage

Figure 10 shows the conversion process for a low input voltage ( $\approx 0.5V$ ).

- **Sampling Phase:** When  $SMPL$  is high, the sampled voltage ( $V_{SAH}$ , red trace) tracks the input.
- **Conversion Phase:** The SAR logic initializes the MSB. Since  $V_{SAH} < V_{DAC}$ , the comparator output is Low. The logic discards the MSB and moves to the next bit.
- **Result:** The DAC voltage ( $V_{SAR}$ , cyan trace) steps down successively. The final digital code converges to  $00_8$  (or near minimum).

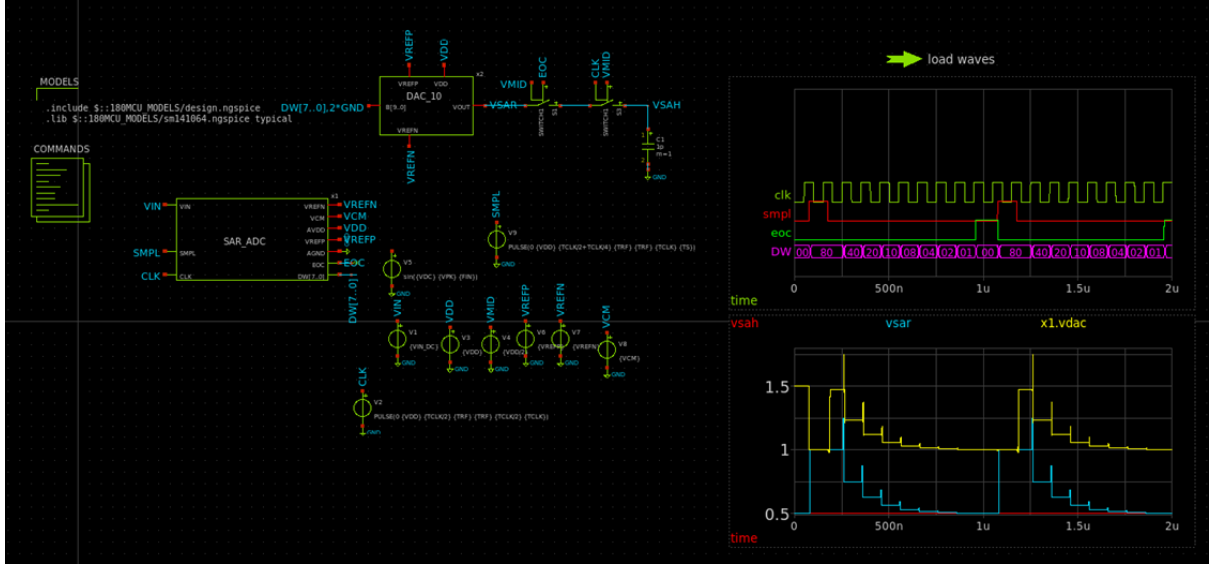


Figure 10: Charge Redistribution for Low Input. The DAC voltage (cyan) steps down to match the low input voltage (red).

## 6.2 Case 2: High Input Voltage

Figure 11 shows the conversion process for a high input voltage ( $\approx 1.5V$ ).

- **Mechanism:** In the first trial,  $V_{SAH} > V_{DAC}$ . The comparator output is High, so the logic keeps the MSB and adds the next bit weight.
- **Result:** The DAC voltage steps up successively. The final digital word ( $DW$ ) reaches  $FF_{16}$  (Maximum Code), indicating the input is at full scale.

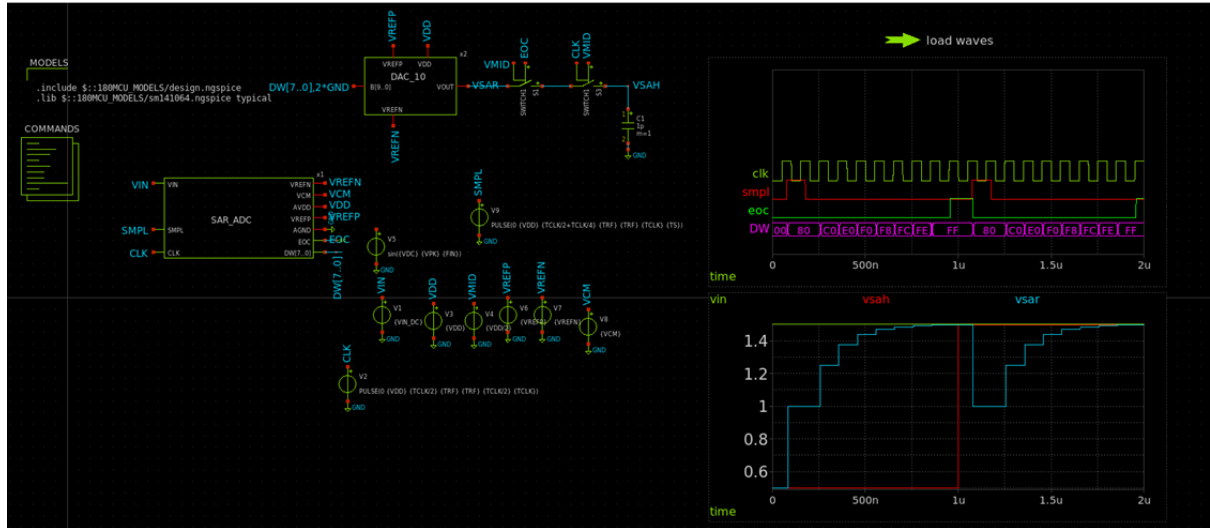


Figure 11: Charge Redistribution for High Input. The DAC voltage steps up to match the high input voltage.

### 6.3 Case 3: Mid-Scale Input Voltage

Figure 12 demonstrates the binary search algorithm for an arbitrary mid-scale input ( $\approx 1.17V$ ).

- **Binary Search:** The waveform clearly shows the "up and down" search nature of the SAR architecture.
- **Sequence:**
  1. Trial 1 (MSB):  $V_{SAR}$  sets to mid-scale.  $V_{SAH} > V_{SAR}$ , so bit is kept (Code  $80_{16}$ ).
  2. Trial 2:  $V_{SAR}$  increases.  $V_{SAH} < V_{SAR}$ , so bit is dropped.
  3. This continues until the LSB is resolved.
- **Result:** The digital output settles at  $AB_{16}$ , and the analog DAC voltage converges perfectly to the sampled input.

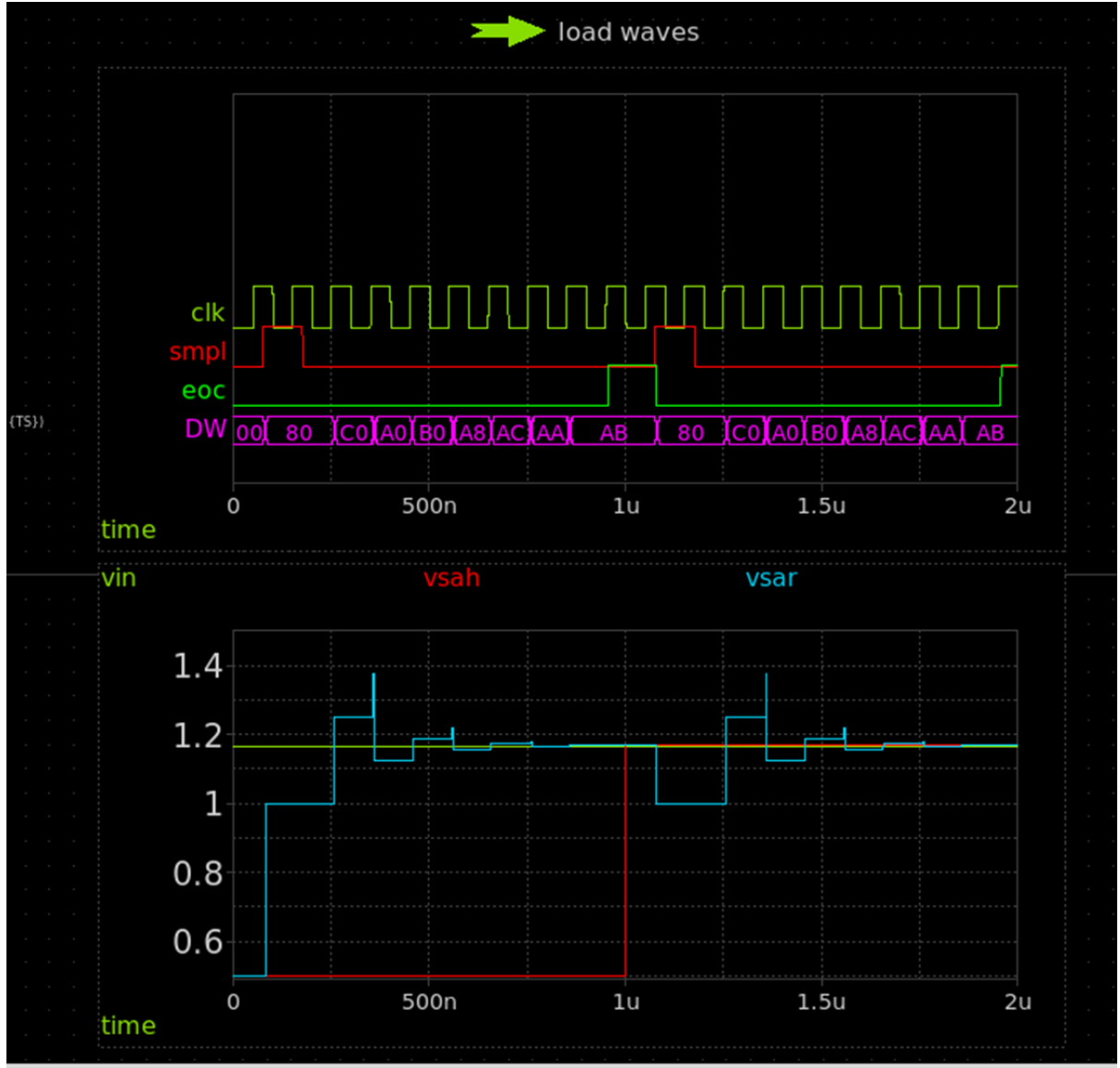


Figure 12: Charge Redistribution for Mid-Scale Input. The system converges to code  $AB_{16}$ .

## System Characterization and Dynamic Analysis

### 7 Transient Analysis

#### 7.1 Signal Tracking

To verify the dynamic behavior of the ADC, a sine wave input was applied. Figure 13 illustrates the operation of the sample-and-hold mechanism.

- The red trace ( $V_{SAH}$ ) tracks the input signal ( $V_{IN}$ , blue) perfectly during the sampling phase.
- It holds the voltage constant during the conversion phase, creating the characteristic "stepped" approximation of the input sine wave.

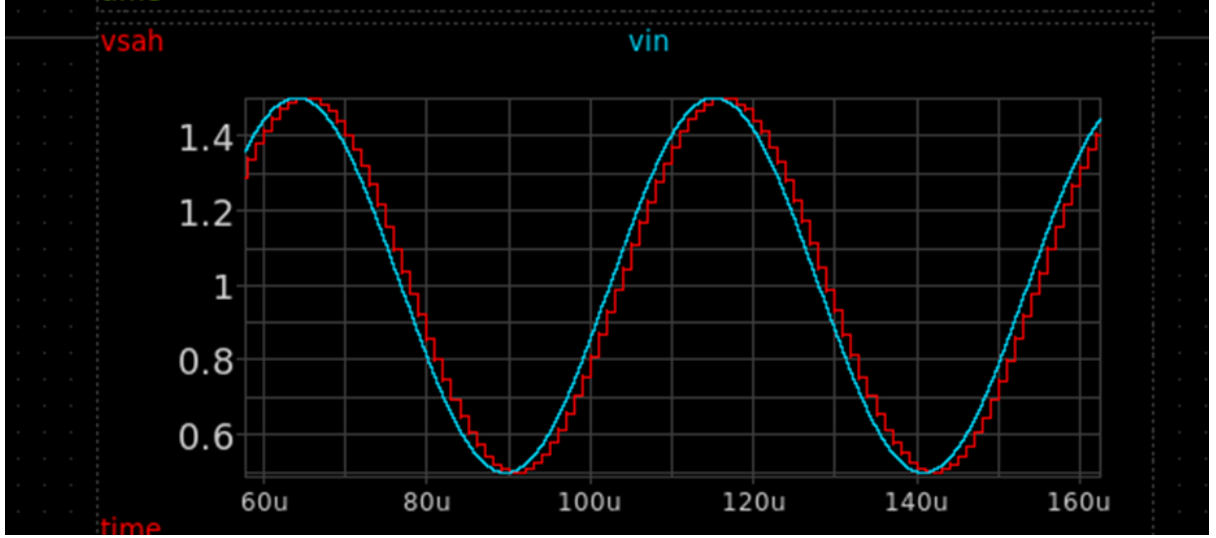


Figure 13: Transient response showing the Sample-and-Hold output (Red) tracking the Input Sine Wave (Blue).

## 7.2 Conversion Detail

Figure ?? presents a zoomed-in view of a single conversion cycle to analyze the internal node behavior.

- **Sampling:** When  $SMPL$  (Red pulse) is High,  $V_{SAH}$  follows  $V_{IN}$ .
- **Bit Trials:** When  $SMPL$  goes Low, the SAR loop activates. The DAC voltage ( $V_{SAR}$ , Cyan trace) performs a binary search. It starts at mid-scale and successively steps up or down to converge toward the held voltage ( $V_{SAH}$ ).

# 8 Dynamic Performance (FFT)

## 8.1 Spectrum Analysis

A Fast Fourier Transform (FFT) was performed on the digital output to quantify the ADC's dynamic linearity. The spectrum (Figure 14) shows a clear fundamental tone with low harmonic distortion.

## 8.2 Performance Metrics

The extracted performance metrics from the simulation are summarized in Table 3. The design achieves an Effective Number of Bits (ENOB) of **7.8 bits**, which indicates excellent performance for an 8-bit architecture, showing that noise and non-linearities are well-controlled.

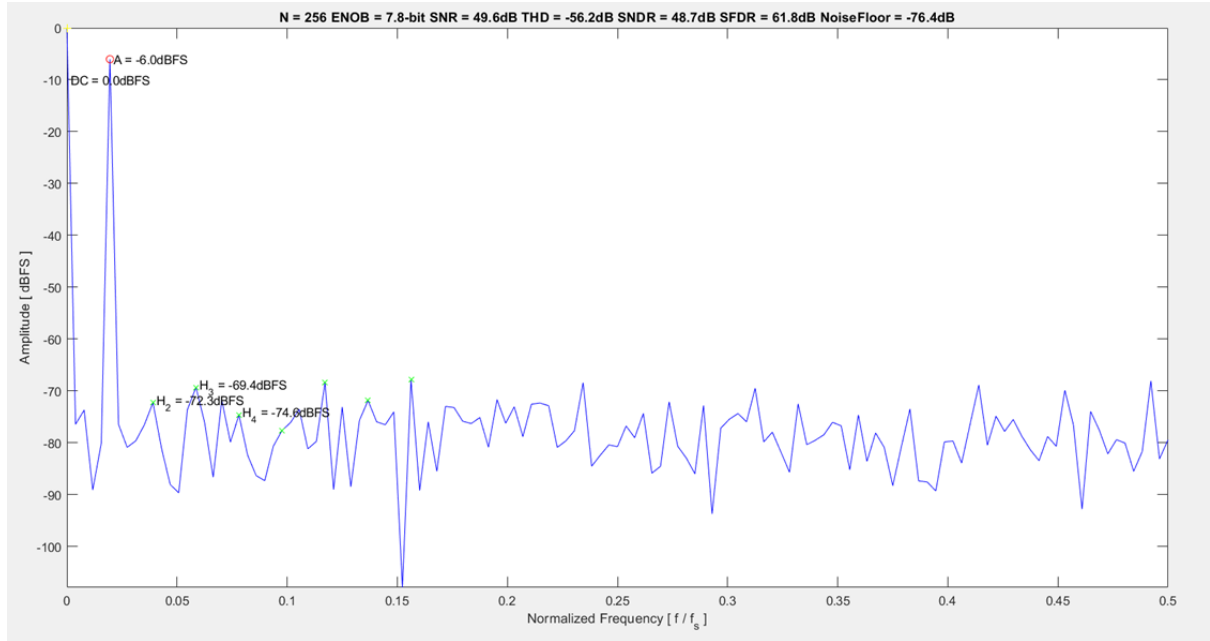


Figure 14: FFT Spectrum of the ADC output with N=256 points.

Table 3: Measured Dynamic Performance Metrics

| Metric                                 | Value           |
|----------------------------------------|-----------------|
| Sampling Points (N)                    | 256             |
| Signal-to-Noise Ratio (SNR)            | 49.6 dB         |
| Signal-to-Noise + Distortion (SNDR)    | 48.7 dB         |
| Spurious-Free Dynamic Range (SFDR)     | 61.8 dB         |
| Total Harmonic Distortion (THD)        | -56.2 dB        |
| Signal Power                           | -6.0 dBFS       |
| DC Power                               | 0 dB            |
| <b>Effective Number of Bits (ENOB)</b> | <b>7.8 bits</b> |